Abstract Window Toolkit (AWT)

Java AWT (Abstract Window Toolkit) is an API to develop Graphical User Interface (GUI) or windows-based applications in Java.

Java AWT Hierarchy

Object

↓

Component

↓

Container

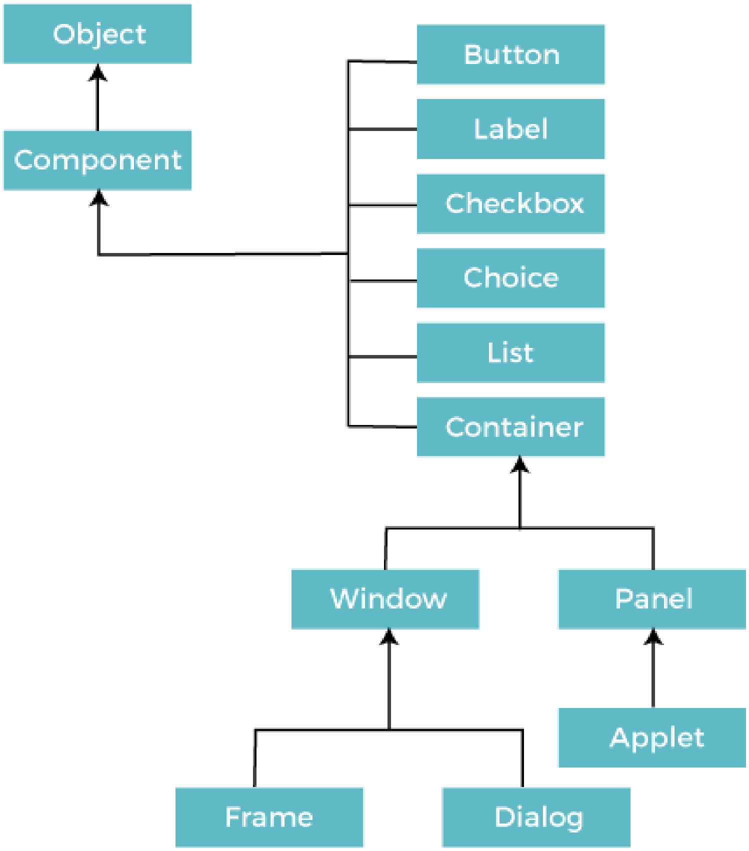
↓

Window

↓

Frame

Class	What it represents	Example	Important features
Object	Root class of Java classes	Object obj	toString(), equals(), hashCode(), getClass()
Component	Basic GUI component	Button, Label, TextField	Can have size, position, visibility, color, font; setSize(), setVisible()
Container	Component that can contain other components	Panel, Window	Can add components using add()
Window	Top-level window without title bar	Window w	Can display GUI components; setVisible()
Frame	Main application window	Frame f	Title bar, border, minimize/maximize/close buttons; setTitle(), setLayout()



Object: Java's predefined root class. Common parent of all Java classes

Component is a subclass of Object. All the elements like the button, text fields, scroll bars, etc. are called components.

Container: The Container is a component in AWT that can contain another component like buttons, text fields, labels etc.

Types of Containers

Window, Panel, Frame and Dialog

Window

The window is the containers that have no borders and menu bars.

You must use frame, dialog or another window for creating a window.

Panel

The Panel is the container that doesn't contain title bar, border or menu bar.

It can have other components like button, text field etc.

Frame

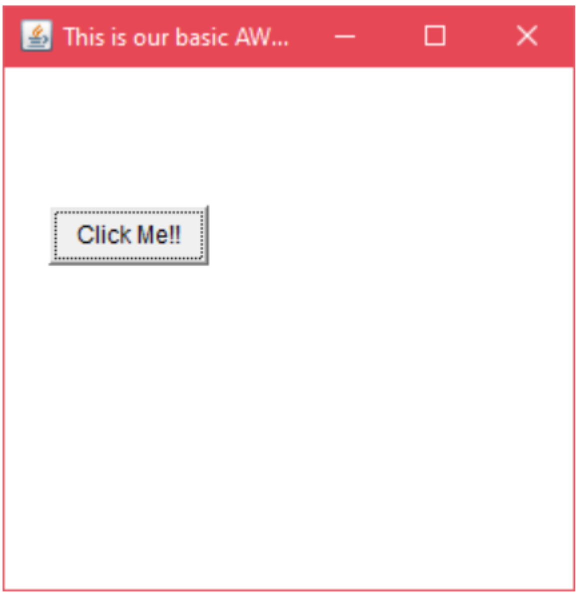
The Frame is the container that contain title bar and border and can have menu bars.

It can have other components like button, text field, scrollbar etc.

There are two ways to create a GUI using Frame in AWT.

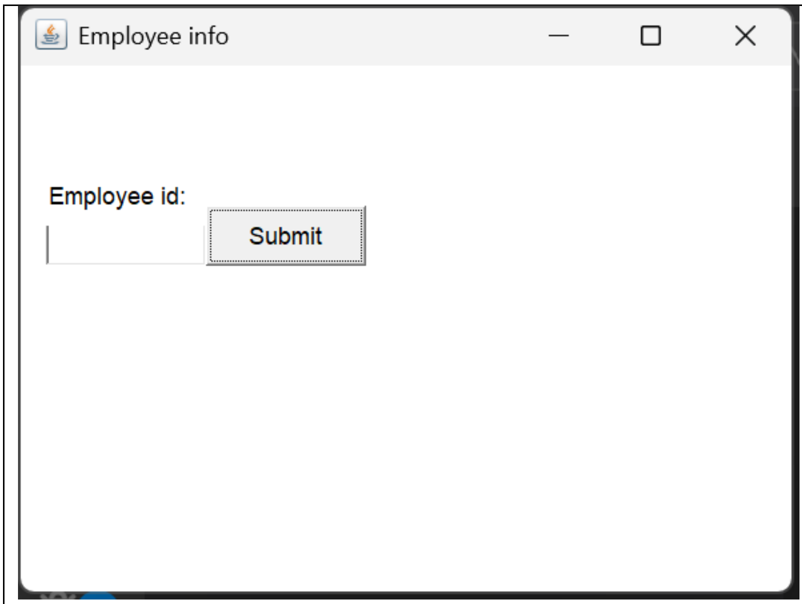
By extending Frame class (inheritance)

```
import java.awt.*;
public class AWTExample1 extends Frame {
    AWTExample1() {
    Button b = new Button("Click Me!!");
    b.setBounds(30,100,80,30);
    add(b);
    setSize(300,300);
    setTitle("This is our basic AWT example");
    setLayout(null);
    setVisible(true);
}
public static void main(String args[]) {
AWTExample1 f = new AWTExample1();
}
}
```



By creating the object of Frame class

```
import java.awt.*;
class AWTExample2 {
    AWTExample2() {
    Frame f = new Frame();
    Label l = new Label("Employee id:");
    Button b = new Button("Submit");
    TextField t = new TextField();
    l.setBounds(20, 80, 80, 30);
    t.setBounds(20, 100, 80, 30);
    b.setBounds(100, 100, 80, 30);
    f.add(b);
    f.add(l);
    f.add(t);
    f.setSize(400,300);
    f.setTitle("Employee info");
    f.setLayout(null);
    f.setVisible(true);
}
public static void main(String args[]) {
AWTExample2 awt_obj = new AWTExample2();
}
}
```

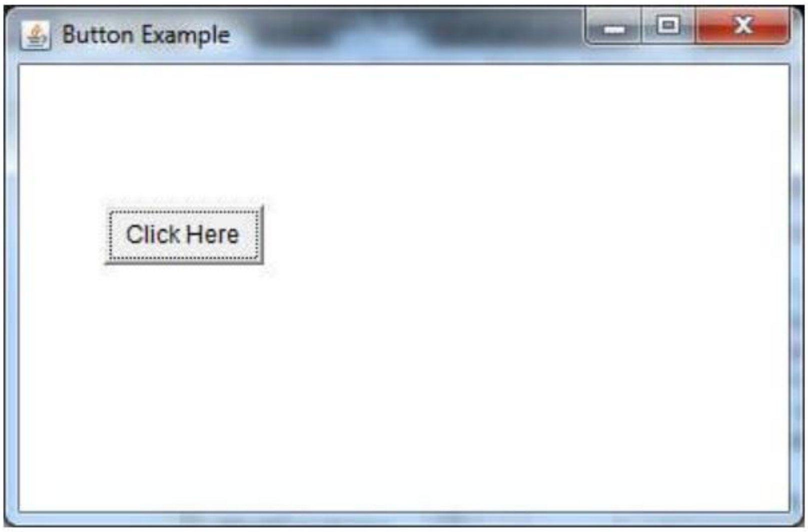


Java AWT Button

Constructor	Meaning
Button()	Button without text
Button("OK")	Button with text **OK**

Creating a Simple Button

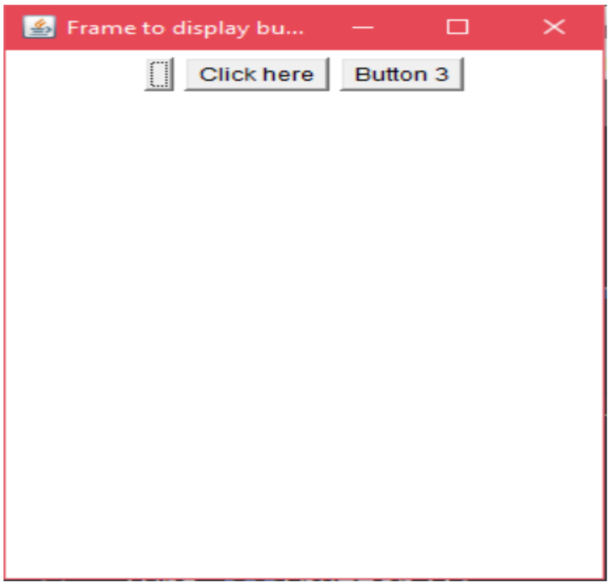
```
import java.awt.*;
public class ButtonExample {
public static void main (String[] args) {
      Frame f = new Frame("Button Example");
      Button b = new Button("Click Here");
    b.setBounds(50,100,80,30);
    f.add(b);
    f.setSize(400,400);
    f.setLayout(null);
    f.setVisible(true);
}
}
```



Creating Multiple Buttons

```
import java.awt.*;
public class ButtonExample2
{
Frame fObj;
Button button1, button2, button3;
ButtonExample2() {
fObj = new Frame ("Frame to display buttons");
button1 = new Button();
button2 = new Button ("Click here");
button3 = new Button();
button3.setLabel("Button 3");
fObj.add(button1);
fObj.add(button2);
fObj.add(button3);
fObj.setLayout(new FlowLayout());
fObj.setSize(300,400);
fObj.setVisible(true);
}
public static void main (String args[])
{
new ButtonExample2();
```

```
}
}
```



Java AWT Label

Constructor	Meaning
Label()	Creates an empty label
Label(String text)	Creates a label with the specified text
Label(String text, int alignment)	Creates a label with text and specified alignment (LEFT, CENTER, or RIGHT)

```
import java.awt.*;
public class LabelExample {
public static void main(String args[]){
    Frame f = new Frame ("Label example");
    Label l1, l2;
    l1 = new Label ("First Label.");
    l2 = new Label ("Second Label.");
    l1.setBounds(50, 100, 100, 30);
    l2.setBounds(50, 150, 100, 30);
    f.add(l1);
    f.add(l2);
    f.setSize(400,400);
    f.setLayout(null);
    f.setVisible(true);
```

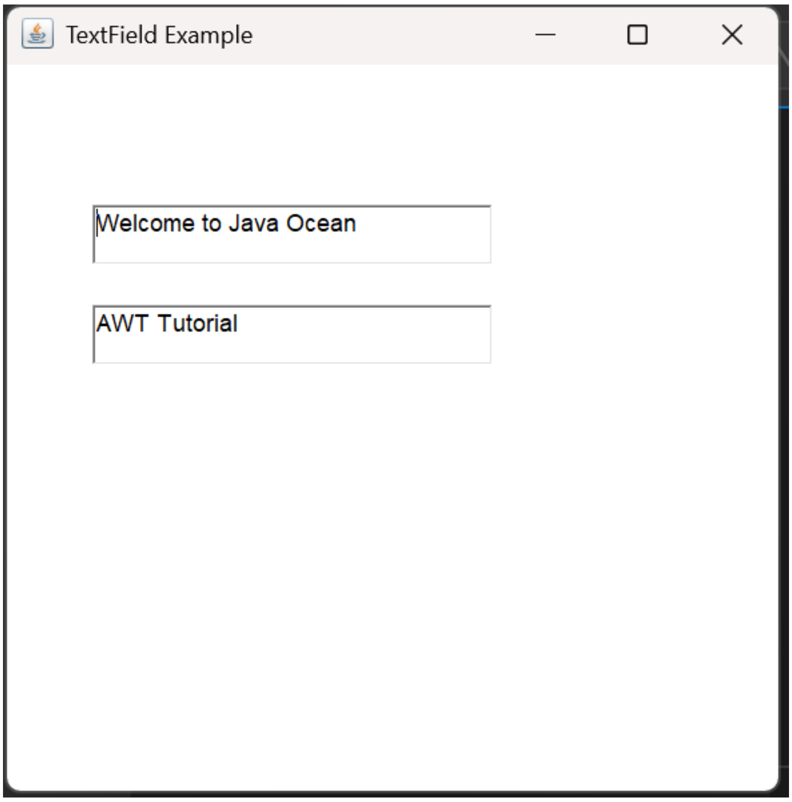
```
}
}
```

Java AWT TextField

Constructor	Meaning
TextField()	Creates an empty text field
TextField(String text)	Creates a text field with the specified text
TextField(int columns)	Creates an empty text field with specified number of columns
TextField(String text, int columns)	Creates a text field with specified text and number of columns

```
import java.awt.*;
public class TextFieldExample1 {
    public static void main(String args[]) {
    Frame f = new Frame("TextField Example");
    TextField t1, t2;
    t1 = new TextField("Welcome to Java Ocean");
    t1.setBounds(50, 100, 200, 30);
    t2 = new TextField("AWT Tutorial");
    t2.setBounds(50, 150, 200, 30);
    f.add(t1);
    f.add(t2);
    f.setSize(400,400);
    f.setLayout(null);
    f.setVisible(true);
```

```
}
}
```

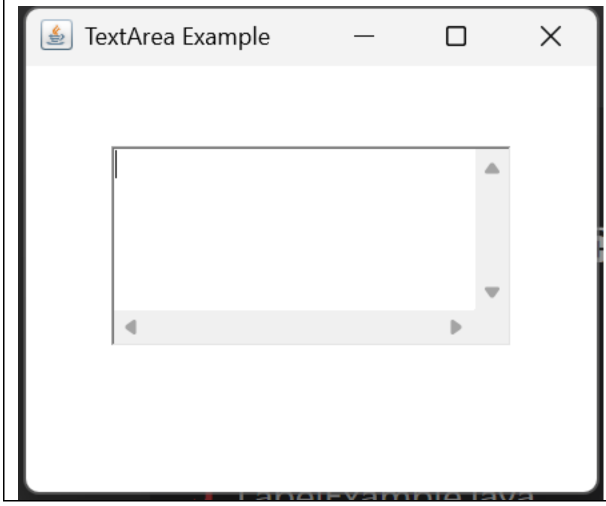


AWT TextArea

Constructor	Meaning
TextArea()	Creates an empty text area
TextArea(String text)	Creates a text area with the specified text
TextArea(int rows, int columns)	Creates an empty text area with specified rows and columns
TextArea(String text, int rows, int columns)	Creates a text area with text, rows, and columns
TextArea(String text, int rows, int columns, int scrollbars)	Creates a text area with text, rows, columns, and scrollbar settings

Constant	Meaning
TextArea.SCROLLBARS_NONE	No scrollbars
TextArea.SCROLLBARS_HORIZONTAL_ONLY	Only horizontal scrollbar
TextArea.SCROLLBARS_VERTICAL_ONLY	Only vertical scrollbar
TextArea.SCROLLBARS_BOTH	Both horizontal and vertical scrollbars

```
import java.awt.*;

public class TextAreaExample {
    public static void main(String[] args) {

        Frame f = new Frame("TextArea Example");

        TextArea t = new TextArea();
        t.setBounds(50, 70, 200, 100);

        f.add(t);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
    }
}
```

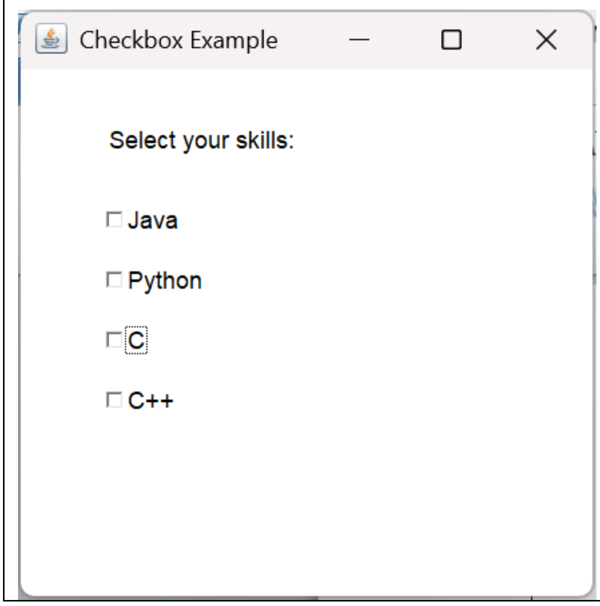


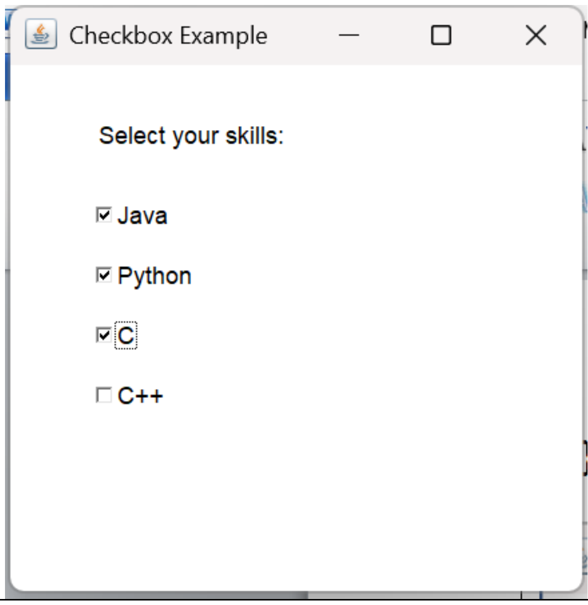
AWT Checkbox

Constructor	Meaning
Checkbox()	Creates an unchecked checkbox without a label
Checkbox(String label)	Creates an unchecked checkbox with the specified label
Checkbox(String label, boolean state)	Creates a checkbox with a label and initial state
Checkbox(String label, CheckboxGroup group, boolean state)	Creates a checkbox with a label, group, and initial state

```
import java.awt.*;

public class CheckboxExample {
    public static void main(String[] args) {

        Frame f = new Frame("Checkbox Example");
```

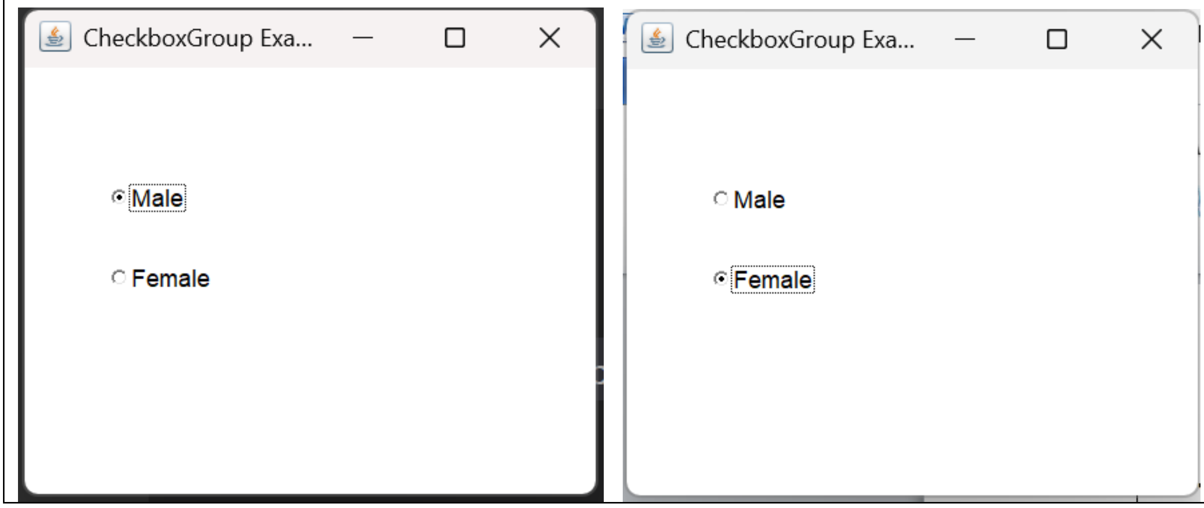
```
        Label l = new Label("Select your skills:");

        Checkbox c1 = new Checkbox("Java");
        Checkbox c2 = new Checkbox("Python");
        Checkbox c3 = new Checkbox("C");
        Checkbox c4 = new Checkbox("C++");

        l.setBounds(50, 50, 150, 30);

        c1.setBounds(50, 90, 100, 30);
        c2.setBounds(50, 120, 100, 30);
        c3.setBounds(50, 150, 100, 30);
        c4.setBounds(50, 180, 100, 30);

        f.add(l);
        f.add(c1);
        f.add(c2);
        f.add(c3);
        f.add(c4);

        f.setSize(300, 300);
        f.setLayout(null);
        f.setVisible(true);
    }
}
```






```
import java.awt.*;

public class CheckboxGroupExample {
    public static void main(String[] args) {

        Frame f = new Frame("CheckboxGroup Example");
```

```
        CheckboxGroup cg = new CheckboxGroup();

        Checkbox male = new Checkbox("Male", cg, true);
        Checkbox female = new Checkbox("Female", cg, false);

        male.setBounds(50, 80, 100, 30);
        female.setBounds(50, 120, 100, 30);

        f.add(male);
        f.add(female);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
    }
}
```

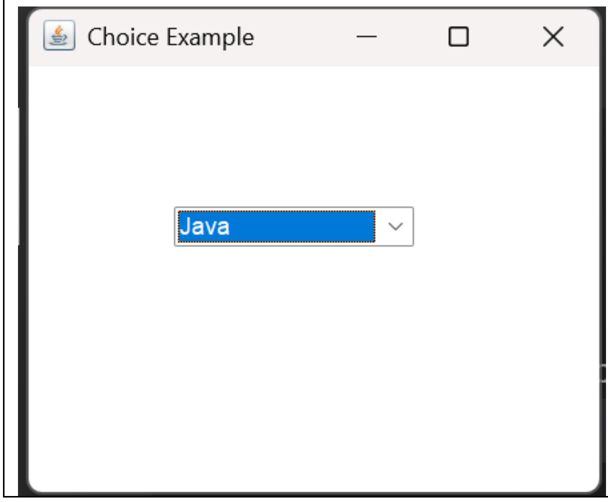


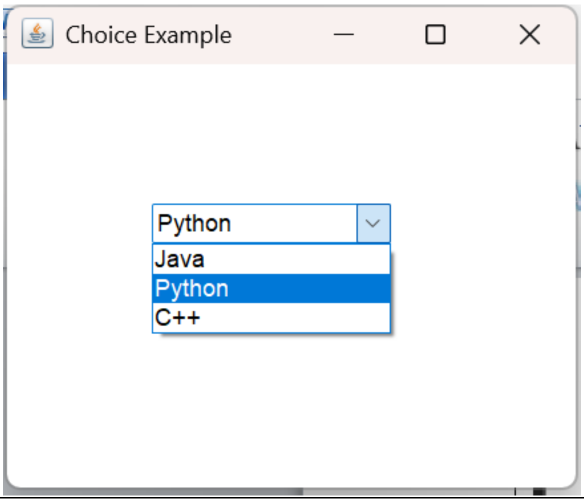
AWT Choice

Constructor	Meaning
Choice()	Creates an empty choice/drop-down list

```
import java.awt.*;

public class ChoiceExample {
    public static void main(String[] args) {

        Frame f = new Frame("Choice Example");

        Choice c = new Choice();

        c.add("Java");
        c.add("Python");
```

```
        c.add("C++");

        c.setBounds(80, 100, 120, 30);

        f.add(c);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
    }
}
```





AWT List

Constructor	Meaning
List()	Creates an empty list
List(int rows)	Creates an empty list with the specified number of visible rows
List(int rows, boolean multipleMode)	Creates a list with specified visible rows and allows/disallows multiple selection

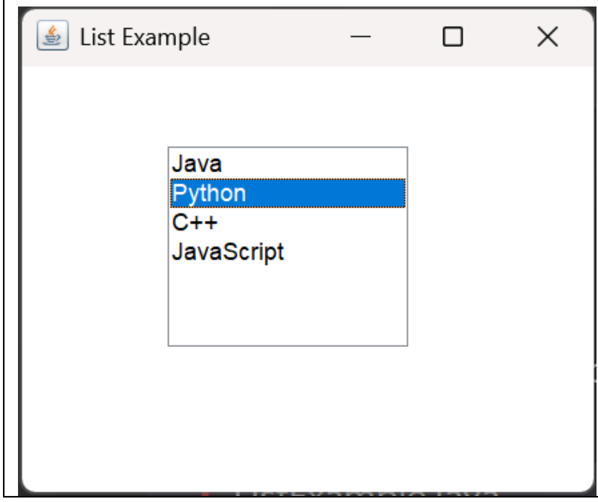
Meaning of multipleMode

- true → Multiple items can be selected.
- false → Only one item can be selected.

```
List l1 = new List(4, true);   // 4 visible rows, multiple selection
List l2 = new List(4, false);  // 4 visible rows, single selection
```

```
import java.awt.*;

public class ListExample {
    public static void main(String[] args) {
```

```
        Frame f = new Frame("List Example");
        List l = new List();
        l.add("Java");
        l.add("Python");
        l.add("C++");
        l.add("JavaScript");
        l.setBounds(80, 70, 120, 100);
        f.add(l);
        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
    }
}
```



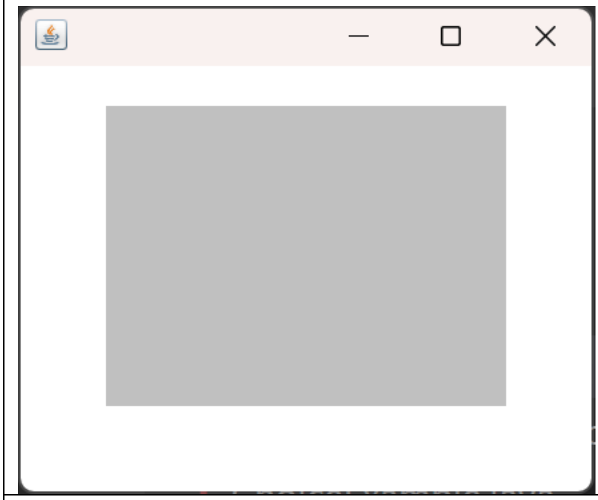
AWT Canvas

Canvas is an AWT component that provides a blank rectangular area where you can draw graphics.

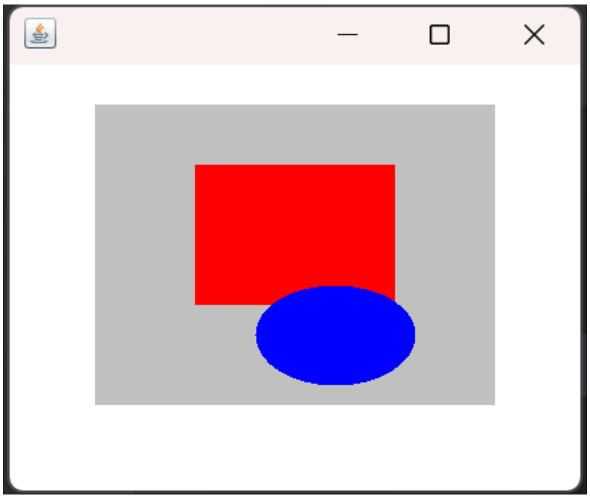
Constructor	Meaning
Canvas()	Creates a new, blank Canvas

```
import java.awt.*;
public class CanvasExample extends Frame {

    CanvasExample() {
        Canvas c = new Canvas();
        c.setBackground(Color.LIGHT_GRAY);
        c.setBounds(50, 50, 200, 150);
        add(c);
        setSize(300, 250);
        setLayout(null);
        setVisible(true);
    }
```

```
    public static void main(String[] args) {
        new CanvasExample();
    }
}
```



```
import java.awt.*;
public class CanvasExample extends Frame {

    CanvasExample() {
        Canvas c = new Canvas() {
            public void paint(Graphics g) {

                // Red rectangle
                g.setColor(Color.RED);
                g.fillRect(50, 30, 100, 70);

                // Blue oval
                g.setColor(Color.BLUE);
                g.fillOval(80, 90, 80, 50);
            }
        };

        c.setBackground(Color.LIGHT_GRAY);
        c.setBounds(50, 50, 200, 150);

        add(c);

        setSize(300, 250);
        setLayout(null);
        setVisible(true);
    }

    public static void main(String[] args) {

        new CanvasExample();
    }
}
```



paint() is a method of the AWT Component class used to draw or display graphics and text on a component.

paint(Graphics g) can be used to draw on a Frame too.

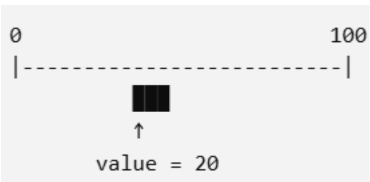
Graphics g → The Graphics object used to draw shapes, text, lines, etc.

AWT Scrollbar

Constructor	Meaning
Scrollbar()	Creates a vertical scrollbar with default values
Scrollbar(int orientation)	Creates a scrollbar with the specified orientation
Scrollbar(int orientation, int value, int visible, int minimum, int maximum)	Creates a scrollbar with orientation, initial value, visible amount, minimum, and maximum values

```
Scrollbar s = new Scrollbar(
    Scrollbar.HORIZONTAL, 20, 10, 0, 100
);
```

Parameter	Meaning
orientation	Direction: HORIZONTAL or VERTICAL
value	Initial/current position of the scrollbar
visible	Amount of the content currently visible
minimum	Minimum value of the scrollbar
maximum	Maximum value of the scrollbar



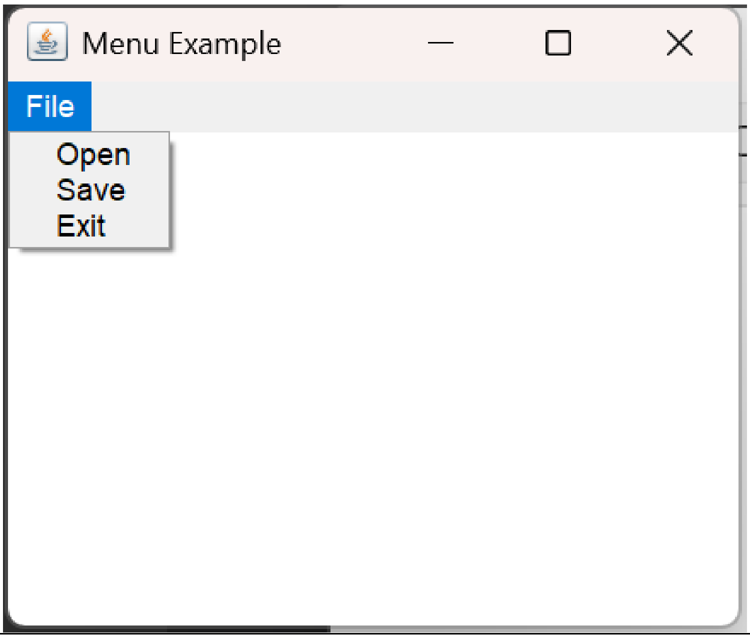
```
import java.awt.*;
public class ScrollbarExample extends Frame {

    ScrollbarExample() {
        // Horizontal scrollbar
        Scrollbar s = new Scrollbar(
            Scrollbar.HORIZONTAL,
            20,     // initial value
            10,     // scrollbar visible amount
            0,      // minimum
            100     // maximum
        );
        s.setBounds(50, 80, 200, 20);
        add(s);
        setSize(300, 200);
        setLayout(null);
        setVisible(true);
    }
    public static void main(String[] args) {
        new ScrollbarExample();
    }
}
```

AWT MenuItem & Menu

Constructor	Meaning
MenuItem()	Creates a menu item without a label
MenuItem(String label)	Creates a menu item with the specified label
MenuItem(String label, MenuShortcut shortcut)	Creates a menu item with a label and keyboard shortcut

```
import java.awt.*;
public class MenuExample {
```

```
    public static void main(String[] args) {
        Frame f = new Frame("Menu Example");
        MenuBar mb = new MenuBar();
        Menu file = new Menu("File");
        MenuItem open = new MenuItem("Open");
        MenuItem save = new MenuItem("Save");
        MenuItem exit = new MenuItem("Exit");

        file.add(open);
        file.add(save);
        file.add(exit);

        mb.add(file);
        f.setMenuBar(mb);
        f.setSize(300, 250);
        f.setVisible(true);
    }
}
```



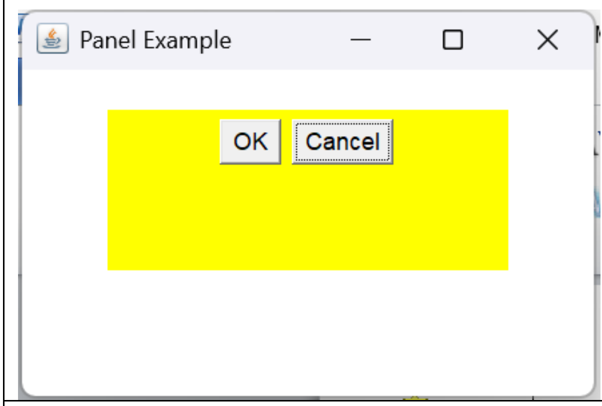
AWT PopupMenu

```
import java.awt.*;

public class PopupExample {
    public static void main(String[] args) {

        Frame f = new Frame("Popup Example");

        PopupMenu pm = new PopupMenu();

        MenuItem copy = new MenuItem("Copy");
        MenuItem paste = new MenuItem("Paste");

        pm.add(copy);
```

```
        pm.add(paste);

        f.add(pm);

        f.setSize(300, 250);
        f.setVisible(true);
    }
}
```

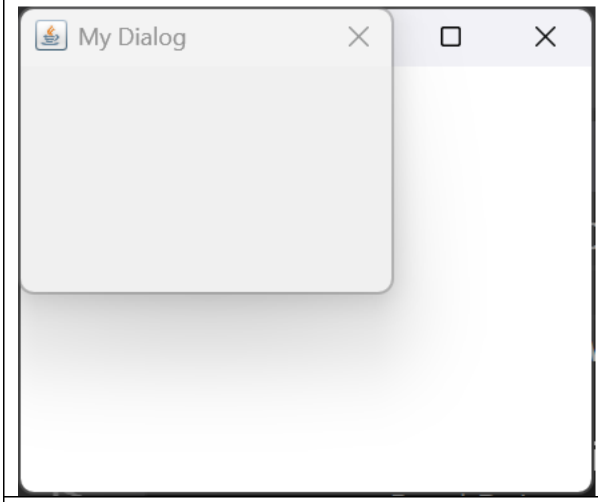
AWT Panel

```
import java.awt.*;

public class PanelExample {
    public static void main(String[] args) {

        Frame f = new Frame("Panel Example");

        f.setLayout(null);   // Manual positioning

        Panel p = new Panel();

        p.setBackground(Color.YELLOW);

        // Set position and size of Panel
        p.setBounds(50, 50, 200, 80);

        Button b1 = new Button("OK");
        Button b2 = new Button("Cancel");

        p.add(b1);
        p.add(b2);

        f.add(p);

        f.setSize(300, 200);
        f.setVisible(true);
    }
}
```



AWT Dialog

```
import java.awt.*;

public class DialogExample {
    public static void main(String[] args) {

        Frame f = new Frame("Main Window");

        Dialog d = new Dialog(f, "My Dialog");

        d.setSize(200, 150);
        d.setVisible(true);

        f.setSize(300, 250);
        f.setVisible(true);
    }
}
```



AWT Toolkit

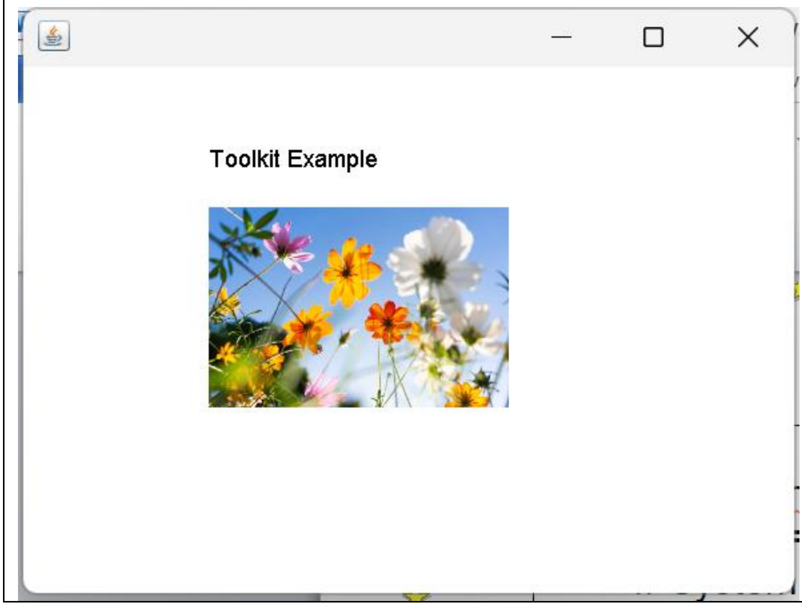
Toolkit is an AWT class that provides access to system-level resources.
For example, using Toolkit, you can get:

- Screen size
- Mouse-related information

```
import java.awt.*;

public class ToolkitExample {
    public static void main(String[] args) {

        Toolkit t = Toolkit.getDefaultToolkit();

        Dimension d = t.getScreenSize();

        System.out.println("Screen Width  : " + d.width);
        System.out.println("Screen Height : " + d.height);
    }
```

```
}

Screen Width  : 1280
Screen Height : 720
```

```java
import java.awt.*;
public class ToolkitExample extends Frame {
    Image img;
    ToolkitExample() {
        Toolkit t = Toolkit.getDefaultToolkit();
        // System beep
        t.beep();
        // Load image
        img = t.getImage("flower.png");
        setSize(400, 300);
        setVisible(true);
    }
    public void paint(Graphics g) {
        g.drawString("Toolkit Example", 100, 80);
        g.drawImage(img, 100, 100, 150, 100, this);
    }
    public static void main(String[] args) {
        new ToolkitExample();
    }
}
```



AWT Component	Simple purpose
Button	Performs an action
Label	Displays text
TextField	Single-line input
TextArea	Multi-line input
Checkbox	Select/deselect option
CheckboxGroup	Select one option from a group
Choice	Drop-down list

List	List of items
Canvas	Drawing area
Scrollbar	Scroll/value control
Menu	Contains menu items
MenuItem	Individual menu option
PopupMenu	Popup/context menu
Panel	Container for components
Dialog	Secondary window
Toolkit	System-dependent features such as screen size, images, and the system beep.